UNDERWATER GARBAGE DETECTION USING COMPUTER VISION

DAY-09

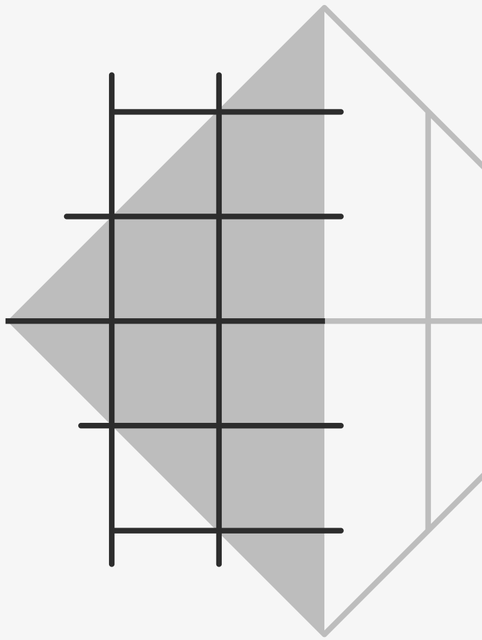
Kavya Hanishka

YOLO11n – 50 Epochs

- Precision: ~0.25
- Recall: ~0.188
- mAP@50: 0.186
- mAP@50–95: 0.119

YOLO11n – ~92 Epochs

- Precision: 0.321
- Recall: 0.199
- mAP@50: 0.185
- mAP@50–95: 0.112

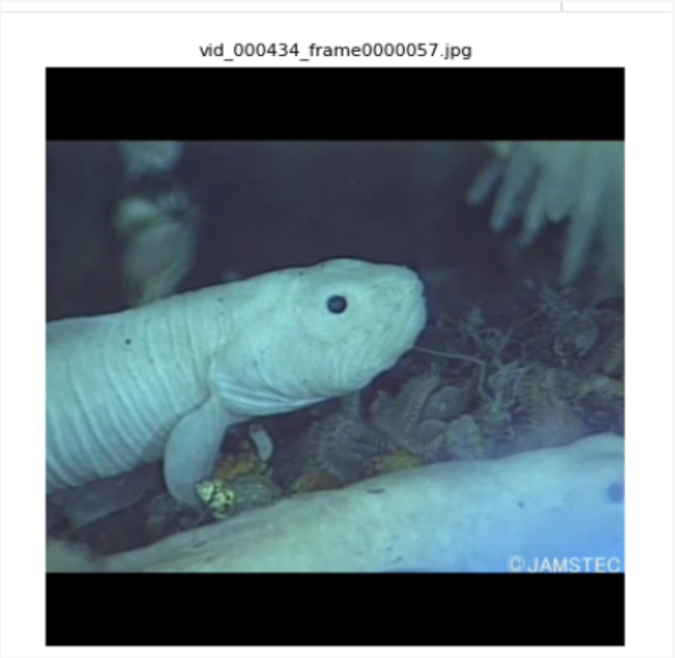


Class wise Analysis

- Precision increased significantly with longer training.
- Recall improved only marginally.
- mAP@50 remained almost unchanged.
- mAP@50–95 decreased slightly.
- More epochs alone did not provide a clear overall performance improvement.

Next Experiment

YOLO11s: Test whether a larger model improves detection performance while keeping the same Dataset V1 split and 640×640 input size.



model with 90 epochs it didn't even predict anything



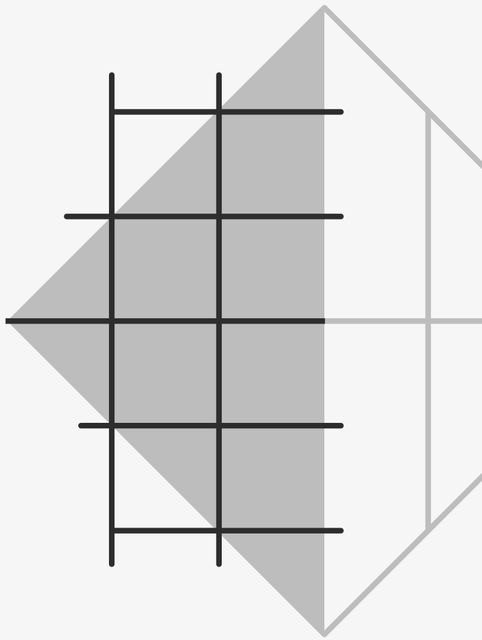
model trained with 50 epochs gave rov for this image

YOLO11S – 100 Epochs

- Precision: 0.394
- Recall: 0.198
- mAP@50: 0.185
- mAP@50–95: 0.126

YOLO11S – 100 Epochs(LR Refinement lr0 = 0.0002)

- Precision: 0.393
- Recall: 0.199
- mAP@50: 0.185
- mAP@50–95: 0.126



Better-performing classes

- rov → mAP50 0.546
- animal_eel → 0.508
- animal_crab → 0.416
- trash_wood → 0.491

Weak classes

- plant → 0.006
- animal_starfish → 0.006
- animal_etc → ~0.0004
- trash_paper → 0
- trash_rubber → 0

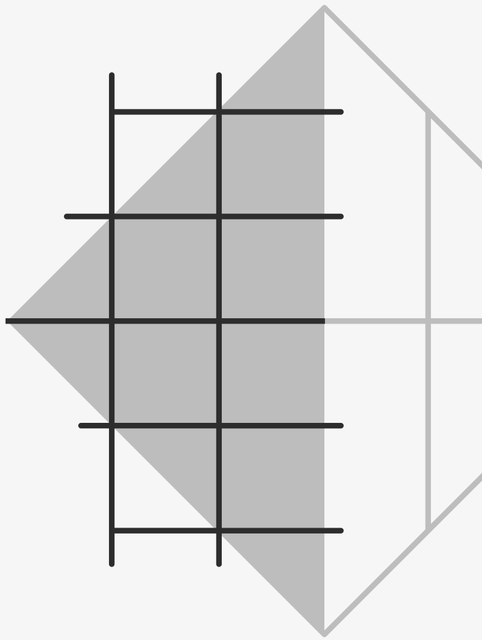
Difficult classes

- plant, animal_starfish, animal_etc
- trash_paper, trash_rubber
- Several rare classes show near-zero detection performance.

This indicates the impact of class imbalance, underwater visibility and small/difficult objects.

Experiment Status — YOLO11s Refinement

- Trial 2 — Mild augmentation: Applied horizontal flip, brightness/contrast, hue/saturation, Gaussian blur and Gaussian noise to training data only.
- Trial 2 status: Training reached ~78/100 epochs before the Kaggle session restart; at epoch 77, it achieved 0.312 Precision, 0.189 Recall, 0.162 mAP@50 and 0.106 mAP@50–95.
- mAP@50–95 decreased slightly.
- More epochs alone did not provide a clear overall performance improvement.



Thank you